Prompt Engineering

Cellula Technologies

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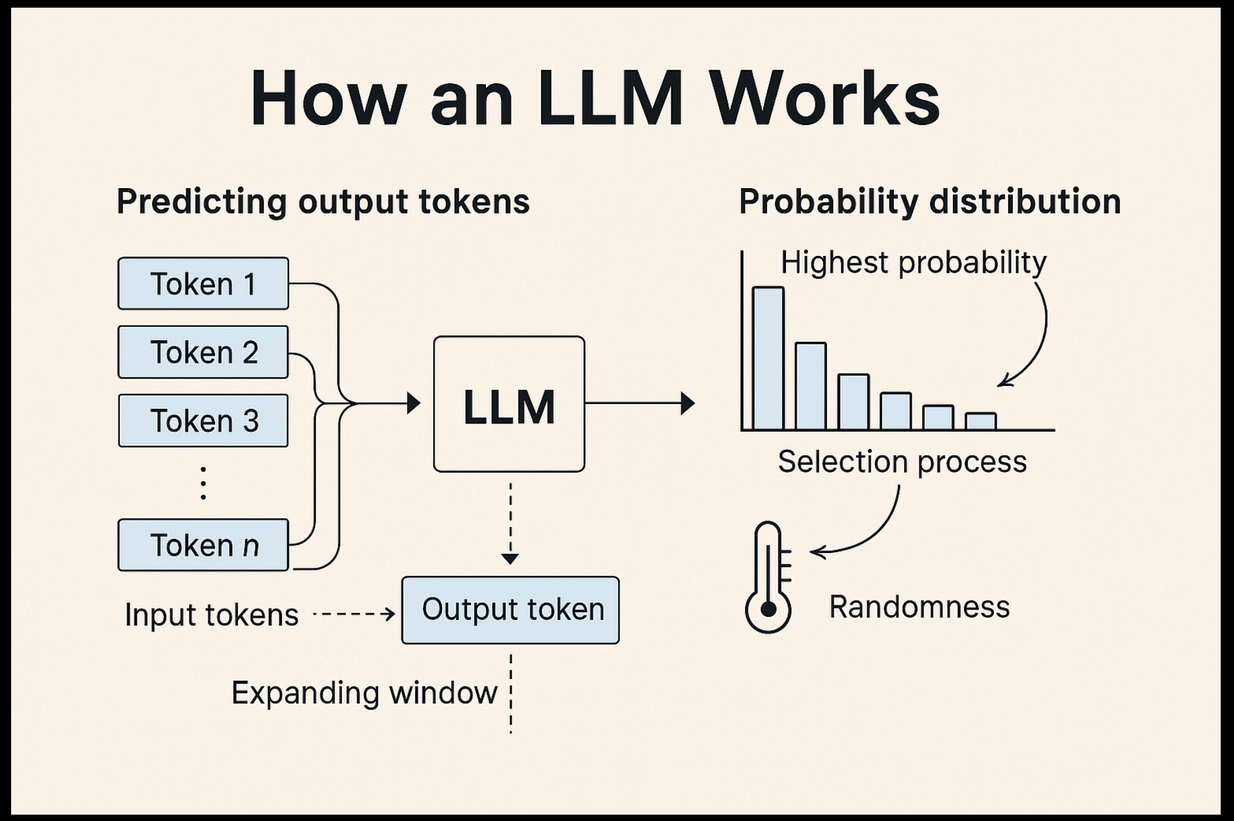
why prompt Engineering ?

- Enhances Human - AI Communication
- Brings accuracy in Results
- Saves time and Efforts
- Improves user Experience



zero shot

Zero-shot prompting means you give the model a task description only, with no examples of how to do it, and it still performs the task correctly.

Prompt:

Classify the sentiment of this sentence as Positive, Negative, or Neutral:
"The movie was painfully slow and boring."

Prompt:

Translate this sentence into Spanish:
"Machine learning is transforming the world."

Prompt:

Extract the names of all cities mentioned in the following text:
"I visited Cairo and Alexandria last summer, then flew to Paris."

Zero-shot struggles when:

- **The task is extremely specific**
- **The output format must be rigid**
- **The task is novel or domain-specific**
- **The instruction is ambiguous**

Zero-shot prompting became especially powerful after the release of large transformer models like: GPT-3 GPT-4
Scaling laws showed that as model size increases, zero-shot performance improves dramatically.

One Shot prompting

Provides a single example, hence the name one shot, the idea is the model has an example it can imitate to best complete the task

Few shot prompting

provides multiple examples to the model, this approach shows the model a pattern that it needs to follow the idea is similar to one-shot but multiple examples of the desired pattern increases the chance the model follow the pattern

Example of one shot

Classify the sentiment of the sentence as Positive or Negative.

Example:

Sentence: "This movie was terrible."

Sentiment: Negative

Now classify:

Sentence: "I love this phone."

Sentiment:

Example of Few shot

Classify the sentiment as Positive or Negative.
Sentence: "This movie was terrible."
Sentiment: Negative
Sentence: "The service was amazing."
Sentiment: Positive
Sentence: "I hate waiting in line."
Sentiment: Negative
Now classify:
Sentence: "I love this phone."
Sentiment:

System prompting

system prompting is Giving the LLM an instruction of system level that defines Tone and style, Constraints, Priority

Example :

“ you are an Expert AI Tutor for Telecommunication engineering,always explain topics,step by step, include equations and digrams when possible, and avoid informal slang

user prompt :

“Explain modulation in communication system”

llm output :

modulation is the process of varying a carrier signal to encode information, lets start with the mathematical representation

System prompting

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Example :

“ you are a humorous pirate, Answer all questions in pirate speak”

user prompt :

“Explain modulation in communication system”

llm output :

Arrr! modulation be the art O’bendin’ a signal to carry yer treasure map o’data!

```python
from langchain_core.messages import SystemMessage

message = SystemMessage("you are Good assistant")

print(message)
```

```
content='you are Good assistant' additional_kwargs={} response_metadata={}
```

Rephrase and Respond (RaR)

Restate the request clearly in your own words, then provide the response based on that restated request

RaR can fail if the user's intent is fundamentally unclear

Rereading prompting (RE2)

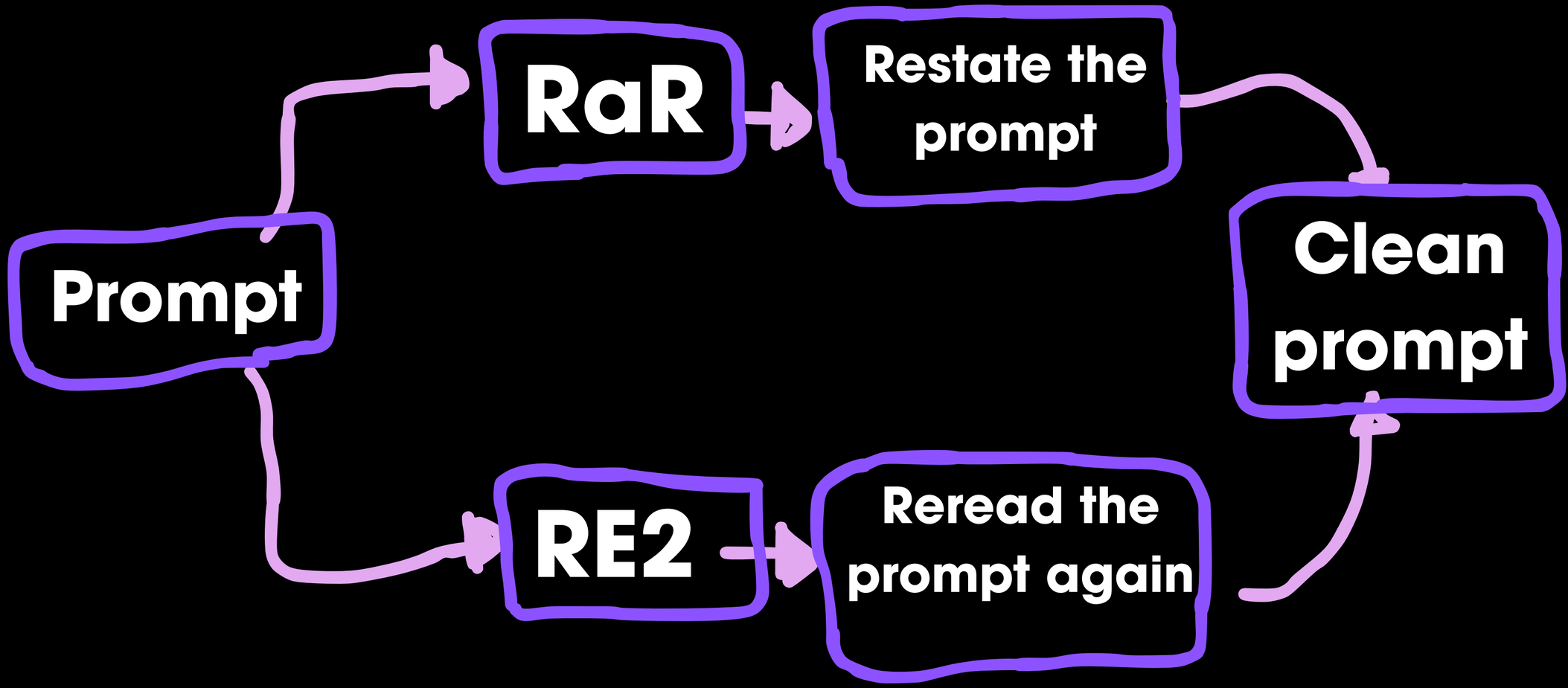
RE2 prompting is a technique where you explicitly instruct
the model to read the entire prompt again carefully
before generating the final answer

Before answering, re-read all instructions carefully then respond

“Reread the entire prompt carefully before answeing”

“before answering, review all instructions again”

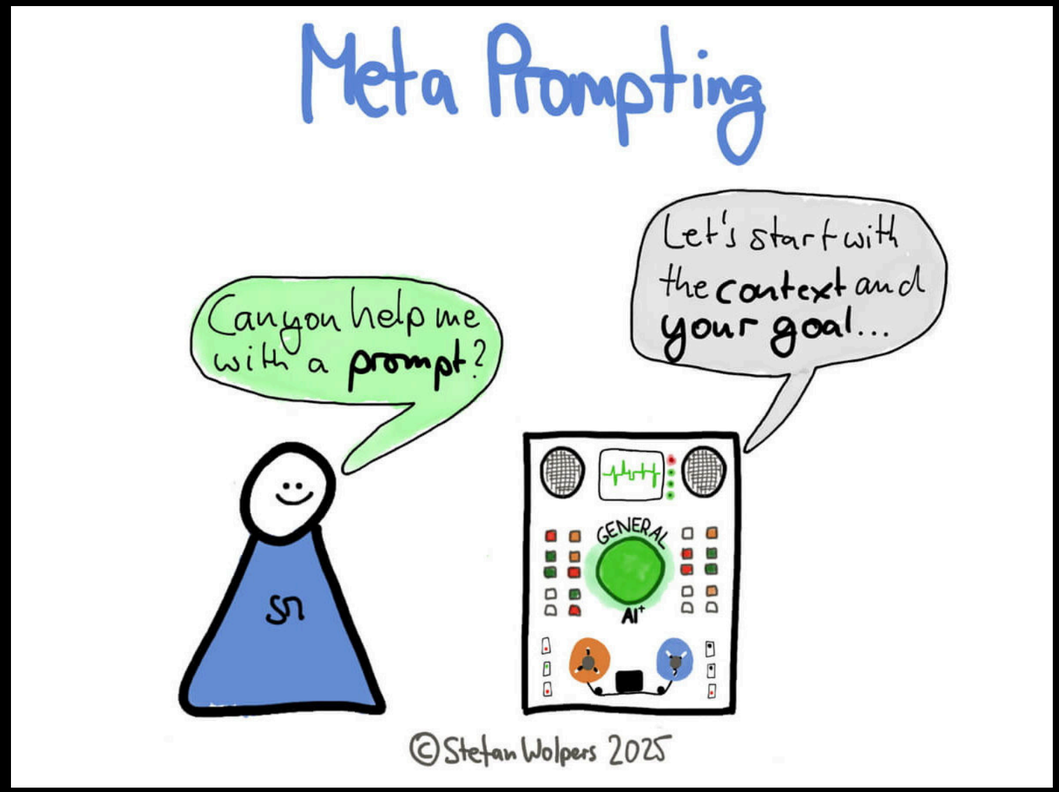
so both RaR and RE2 do similar Job



Meta prompting

Higher level promptng technique used with LLMs where you give the model instructions about how to generate or Improves its own reasoning strategy, rather than asking it to answer a Question

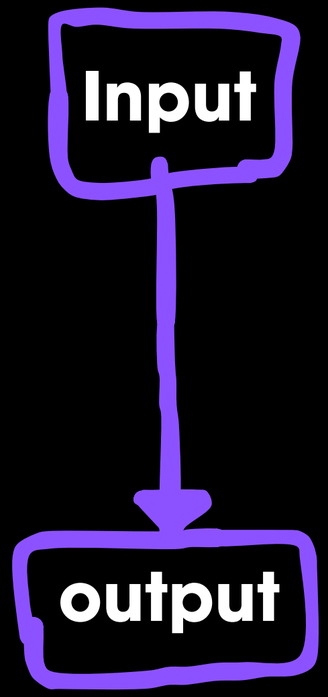
it is prompting about prompting

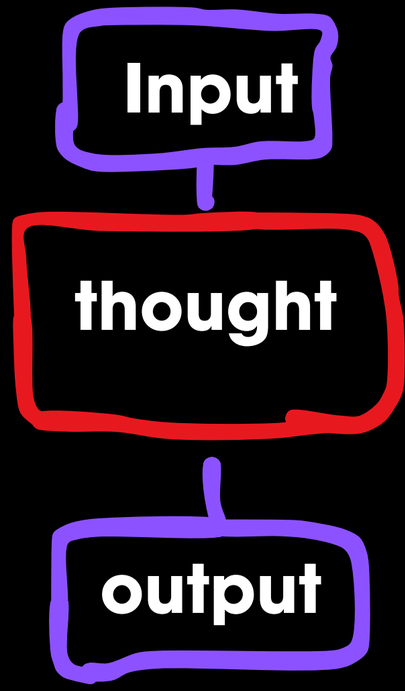


Chain of Thought

chain of thought is a technique where you ask the model to reason step by step instead of jumping final answer

without COT, models may do some hidden reasoning internally but provide just the output, COT makes the reasoning explicit





The LLM response includes the chain of thought reasoning which means more output tokens which means more money and take longer

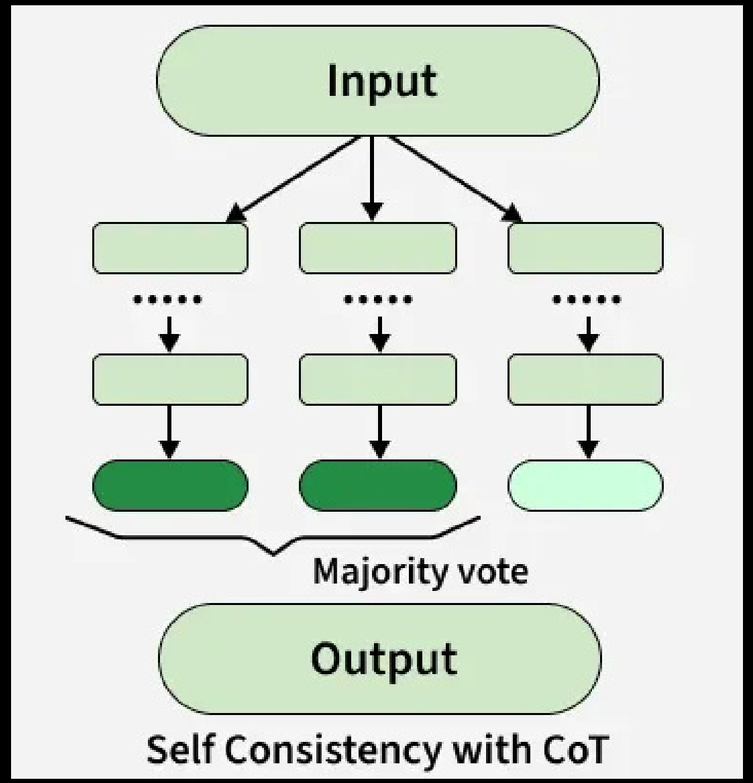
wrong chain is often worse than no chain because it is wrong and waste of money

Self Consistency

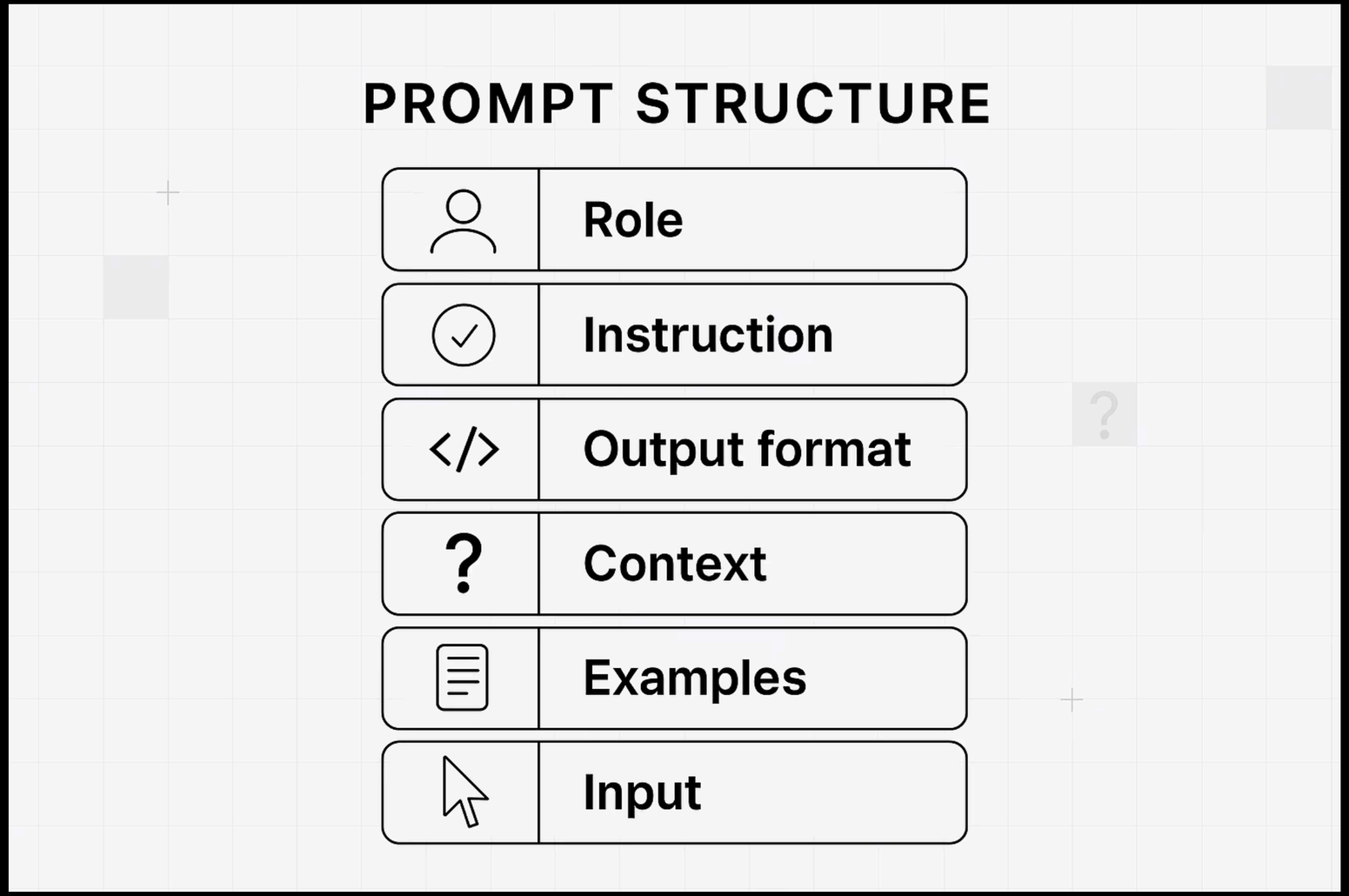
Instead of trusting one reasoning path, generate multiple reasoning chains and aggregate their final answers

one of the most common methods is by using different Temperature values

or you can use different prompt for each chain



Prompt structure



Thank You

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